Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
3. HT	(a)			Any 2 × (1) from Same size/shape rod (or reference to length, gauge, thick(ness)) MUST GIVE PROPERTY (1) Heated from same start point / rod ends must be touching / with same flame (1) DO NOT Accept heated by the same amount Same {volume /mass/ amount} of Vaseline (1) Same starting temperature for materials/ heated from room temperature (1) Same {type/mass/weight} of pin MUST GIVE PROPERTY(1) Drawing pin placed equal distance from flame / drawing pin placed at end of rod (1) Accept: repeats (1)	2			2		2
	(b)			Copper is the best (thermal) conductor (1) so should drop off quickest (1) But it didn't, so not as expected (1) OR According to their results the order of conductivity is aluminium, copper, brass, iron (1) Theoretical values order them copper, aluminium, brass, iron (1) So not as expected since true order not obtained / So brass and iron as expected but aluminium and copper not as expected (1) For 2 marks: Iron is the poorest (thermal) conductor (1) so should drop off last (1) it did, so as expected – 0 marks For 1 mark: The higher the (thermal) conductivity the shorter the time to drop off (1) Allow converse Do not credit 'not as expected' alone			3	3		3

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
					Marks Available					
	(c)			Ionic bonding/ionic lattice/ionic stucture/strong bonds between (+ and -) ions (1) ions are free to move (when liquid / molten) / bonds are broken {when liquid/molten}(1) Insulator when solid / conductor when {liquid/molten} (1) Do not accept electrons for ions	3			3		
				Question 3 total	**5**		**3**	**8**		**5**